

Safe Paths to School: Evidence from Bogotá’s Walking to School Caravans*

Rafael Hernández-Pachón[†], Isabela Munevar[‡]

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Abstract

In many cities in low- and middle-income countries, parents prioritize proximity to schools but face significant coordination challenges and safety concerns regarding their children’s commute. This paper examines the impact of *Ciempiés*, a Walking to School Bus program in Bogotá, Colombia, which transports children to and from school via supervised walking caravans. Using a differences-in-differences model and administrative data from 2016-2024, we exploit the staggered rollout of the program across public schools to estimate causal effects on educational outcomes and road safety. We first analyze the program’s impact on children’s educational trajectories using individual-level data, comparing different treatment cohorts with not-yet-treated cohorts at various points in time relative to the program start. We find substantial reductions in school switching (7.7 percentage points) and switching between school branches (10.6 percentage points) after two years, representing 53% and 45% decreases from baseline rates, respectively. We also observe suggestive evidence of reduced grade repetition and dropout, though these effects are imprecisely estimated. We then examine the program’s impact on traffic accidents involving primary school-age children by comparing census tracts that overlap with school paths in treated versus not-yet-treated areas. We find no significant effects on child traffic accidents, suggesting the program’s safety benefits may be offset by reduced adult supervision relative to individual family transport in the absence of the program. At USD 285 per child per year, *Ciempiés* represents a cost-effective intervention that strengthens children’s attachment to school by offering a reliable transportation mode for families, offering valuable lessons for increasing education access and daily school attendance in urban settings across low- and middle-income countries.

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[†]University of Zurich; Email: rafael.hernandez@econ.uzh.ch

[‡]Universidad de los Andes, Chile

1 Introduction

School attendance is a prerequisite for learning, yet many children in Low- and Middle-Income Countries (LMICs) continue to miss school regularly due to lack of motivation, high costs of attending school (distance, transportation constraints, safety concerns, among others), or parental constraints. In urban settings across Latin America, parents face challenges that hinder their children’s consistent attendance. They must juggle work responsibilities with caregiving duties (which include sending or accompanying children to school), navigate safety concerns about their children’s commute, and coordinate transportation in contexts where formal school transport systems are often inadequate or nonexistent.

The COVID-19 pandemic had a profound and lasting impact on educational attendance patterns. School attendance fell dramatically across Latin America, and many families—particularly those with less-educated parents—have struggled to reestablish consistent school routines (Acevedo et al., 2023; Schady et al., 2023). Even as schools reopened (and Latin America was the region with the longest school closures in the world), the disruption to parent’s daily coordination efforts has left many children more vulnerable to absenteeism.

For working parents, ensuring their children get to school safely can become a hurdle to overcome every day. In many urban areas of LMICs, children must walk long distances to school, often through dangerous or heavy-traffic neighborhoods. Parents face difficult trade-offs: they can accompany their children to school but miss hours of work, send their children alone but worry about their safety, or keep their children home when coordinating school transportation fails. These daily decisions about school attendance are shaped not only by parents’ valuation of education but also by the practical constraints they face when balancing work and family responsibilities.

The consequences of these hurdles that hinder attendance extend beyond individual families. Irregular attendance undermines learning outcomes (Aucejo and Romano, 2016; Liu et al., 2021), increases dropout risk (Koppensteiner and Menezes, 2021), and may affect peers (Goodman, 2014), especially when students coordinate their absences (Bennett and Bergman, 2021). When attendance depends heavily on family coordination efforts, it becomes more sensitive to disruptions (e.g. changes in work schedules, transportation problems, shocks to safety perception). This creates a cycle where children from families facing the greatest logistical or coordination challenges are most likely to miss school, deepening educational inequalities.

Traditional policy responses to faltering attendance have focused on the demand side, making attending school more attractive with programs like Conditional Cash Transfers

(CCT)¹ and school meals². Other demand-side interventions provide parents with information about their children’s attendance or missing assignments (Bergman, 2021; Berlinski et al., 2025; Lichand et al., 2024). These interventions reduce monitoring costs for parents, especially in middle school, when parents and children begin to grow apart and it becomes more difficult for parents to monitor children’s effort. While these approaches can be effective, they often require substantial resources and do not directly address the coordination challenges that working families face. Interventions that reduce the cost and complexity of getting children to school may offer complementary ways to improve attendance and children’s educational trajectories.

This paper evaluates one such intervention: Walking to School Caravans (WSC), which provide supervised walking caravans for primary school children in public schools. In Bogotá’s version of the program, called *Ciempies* (spanish for Centipede) and launched in 2019, trained monitors lead groups of children along predetermined safe paths with designated meeting points where parents or caregivers can drop off and pick up their children. Children walk together to school, the youngest even hand by hand, learning about road safety along the way. By reducing both the time burden of accompanying children to school and appeasing safety concerns about children’s commutes, the program effectively lowers the cost of school attendance for participating families.

The *Ciempies* program is particularly well-suited for studying the coordination efforts to bring children to school because it was implemented through a staggered rollout across Bogotá schools from 2019 to 2024 (and continues to expand to this day), providing quasi-experimental variation in treatment timing. Using rich administrative data from Bogotá’s Secretaries of Education and Mobility, combined with a Difference-in-Differences (DiD) approach, we evaluate the program’s impact on children’s educational outcomes and road safety.

We find that enrollment in *Ciempies* changes children’s educational trajectories. Our main finding is that the program reduces school switching and switching between school branches by 7.7 and 10.6 percentage points, respectively, after two years. This suggests that the program helps children become more attached to their school. We also find reductions of 4 and 5 percentage points, respectively, in grade repetition and dropout after two years of treatment, although these impacts are imprecisely estimated. These effects are substantial,

¹Almost all CCT that focus on education require school enrollment and attendance of 80% or 85% of school days. See Baird et al. (2014), Fiszbein et al. (2009), or Millán et al. (2019) for a review.

²Demand-side interventions contrast with supply-side interventions, like school construction, that bring schools closer to children (e.g., Burde and Linden, 2012; Dinerstein et al., 2023; Duflo, 2001; Khanna, 2022).

especially considering the program’s low cost of USD 285 per child per year in 2023,³ making it a cost-effective alternative to intensive mentoring programs and traditional transportation options, such as school buses or bicycles.

However, we find no evidence that the program improved road safety for children of primary school age. We follow two approaches here. First, we use geocoded traffic accident data to compare census tracts around treated versus not-yet-treated school paths and find that the program has no effect on the number of accidents involving children under 12. Since this data records accidents involving all children in that age range, irrespective of the school they attend, we also examine school-issued alerts regarding accidents on the way to school. We compare the number of alerts issued in treated schools with those issued in not-yet-treated and never-treated schools and find that, if anything, adopting the program increased traffic accidents (although the results are not significantly different from zero).

Despite the program having a road safety component (both in the design of the school paths to minimize traffic and insecurity risk, and in the educational component that promoted safe commutes and observance of traffic signs), our null effects are probably explained by the counteraction of two opposing forces. On the one hand, the program’s road safety component should have decreased traffic accidents, but on the other hand, adult supervision also decreased with the program implementation (since parents would usually accompany their children to school, while now there are only two monitors for up to 50 children).

This study contributes to several strands of literature. First, it adds to the school transportation literature, by providing the first evaluation of a policy that provides safe school transportation at a fraction of the cost of traditional transportation alternatives. This literature has focused mostly on bicycles in LMICs, and school buses in high income countries. Bicycles have been shown to improve girls’ enrollment, test scores, motivation and empowerment (Muralidharan and Prakash 2017; Fiala et al. 2025). In high income countries, busing has increased racial integration but with mixed evidence regarding its effects on learning (Angrist et al. (2022) find no effects on achievement, while Setren (2024) finds positive effects on test scores and dropout). We expand this literature by focusing on Walking to School Buses (WSB), which are cost-effective and can enhance children’s social and language skills by mixing ages in a playful commute.

We also contribute to a booming literature on neighborhood violence and insecurity, and ways to address these city challenges. A big chunk of this literature has focused on

³For reference, the monthly minimum wage in Colombia was USD 240 in 2023. Additionally, according to Colombia’s 2024 Quality of Life Survey (QLS), parents who hire a transportation service for their children in public schools pay around USD 34 per month.

sexual street harassment and domestic violence (e.g., [Amaral et al., 2025](#); [Borker, 2021](#); [Amaral et al., 2021](#)). The evidence regarding road safety is scarce.⁴ We are the first paper to examine a policy designed to improve commuter safety for children.⁵ Furthermore, we show that a community-based solution to irregular attendance in LMICs, that addresses coordination challenges of parents to bring their children to school, can boost attendance and improve school trajectories of children.

More broadly, this project also adds to the growing literature on neighborhood effects and place-based policies aimed at helping households living in economically disadvantaged neighborhoods ([Chetty et al. 2016](#); [Chetty and Hendren 2018a](#); [Chetty and Hendren 2018b](#)). We contribute to this literature by showing that intervening neighborhoods and creating safe paths to school for children can increase their motivation to attend school and improve their educational outcomes, without having to move families or children away from their communities and social networks.

In contexts where many families struggle to balance work and caregiving responsibilities, interventions that reduce the complexity and cost of school attendance can complement traditional education policies or investments in school inputs. The success of *Ciempiés* suggests that relatively simple, community-based approaches can generate substantial educational benefits while improving neighborhood safety, offering a promising model for other urban contexts in LMICs.

The rest of the paper proceeds as follows: Section 2 describes the setting of our study and the data. Section 3 discusses the differences-in-differences methodology, and Section 4 presents results. We conclude in Section 5.

2 Setting and Data

Our study focuses on Bogotá, the capital and largest city in Colombia, whose public school system served an estimated 1,142,000 preschool, primary, and secondary students in 2024. Families in Bogotá can choose between public and private schools, with public school admissions coordinated through a centralized assignment system. In 2024, 63% of students attended public schools, while 37% were enrolled in private institutions. The public school choice system transitioned from a Boston-style mechanism to a Deferred Acceptance algorithm during

⁴Among the exceptions is [Bauernschuster and Rekers \(2022\)](#), who study the effects of speed limit monitoring on traffic accidents.

⁵[Koppensteiner and Menezes \(2021\)](#) study the effect of homicides near schools, children residencies and on the way to school on school attendance and attainment.

2022–2023. Our analysis focuses exclusively on the public sector, as the *Ciempiés* program operates only in public schools.

Parents in Bogotá have numerous public school options, and while distance plays a role in school choice, families do not always choose the nearest public school. This behavior reflects the consideration of multiple factors—such as school quality, perceived safety, and convenience regarding caregivers’ work.⁶ Yet, this choice flexibility is not the same across socioeconomic groups, creating systematic disparities in school commuting patterns.

Figure 2 presents a binscatter regression that reveals a socioeconomic gradient in school distance. Children with lower SISBEN III scores, which indicate higher vulnerability, tend to live farther from their school. Those in extreme poverty (score = 0) travel, on average, 2.2 kilometers to school, while children with higher (though still vulnerable) scores travel around 1.4 kilometers. This pattern suggests that the poorest families face additional barriers to access, potentially due to informal housing in areas with poor public provision of education or late assignment to schools with available slots but in distant locations.⁷

The gradient in school distance translates into concrete burdens for vulnerable families. Figure 5 demonstrates that average travel times to school vary systematically by socioeconomic strata among children who commute by foot or bike. Children from the lowest strata (1 and 2) spend significantly more time commuting than those from higher strata, with strata 1 children requiring nearly 13 minutes on average compared to under 10 minutes for strata 4 children. These differences impose higher time costs and additional coordination challenges on low-income households, making consistent school attendance more challenging to maintain.

The transportation choices available to families reflect both preferences and constraints. Figure 3 illustrates that walking remains the most common way to commute to school in Bogotá, accounting for 66.4% of all school trips. This pattern is more pronounced in public schools, where students are significantly less likely to use paid transportation than their counterparts in private schools. The data show that only 16.4% of students use school buses, 9.4% rely on public transport, 4.7% travel by private vehicle, and 3.0% use bicycles.

For families who do pay for school transportation, the financial burden is substantial. Figure 4 presents the distribution of monthly transportation expenditures among households

⁶Many families, especially those with young children, enroll their children in schools close to their workplaces to facilitate drop-off and pick-up.

⁷While in many LMICs richer families live farther from schools due to suburban housing patterns (Beuermann et al., 2023; Hofflinger et al., 2020), the inverse gradient in Bogotá reflects the concentration of urban poverty in poorly connected peripheries and the challenges of navigating the centralized assignment system.

that report paying for school transportation services. On average, the amount that families pay per month for transportation services is approximately COP 170,000 (roughly USD 43). This amount is nearly double the cost that the city pays per child per month in *Ciempiés* (USD 23.75), highlighting the program’s cost-effectiveness relative to private transportation alternatives.

The pattern that emerges is that many public school students live relatively far from school but cannot afford paid transportation options. This creates a challenge for low-income families, who must either invest significant time accompanying children on long walks to school or find alternative arrangements that may compromise safety or reliability.

In sum, while Bogotá offers choice within its public education system, geographic, logistical, and financial frictions constrain that choice for the poorest families. The consequences of this are longer travel times, greater need for coordination, and limited use of formal transportation services. This can lead to inconsistent school attendance and exacerbate inequality in access to education. The *Ciempiés* program operates within this context, attempting to address the coordination problem parents face in bringing their children to school every day by providing a community-based transportation solution.

2.1 The Ciempiés Program

Bogotá’s Secretaries of Education and Mobility launched *Ciempiés* in 2019. The program was designed to address safety concerns and coordination challenges faced by families whose children walk to school, particularly in unsafe areas and with high incidence of traffic.

The program operates through supervised walking caravans that transport children to and from school along predetermined safe routes. In each participating school, program staff map out an optimal walking path that avoids crime hotspots, minimizes exposure to heavy traffic, and connects the school with the highest concentration of children who walk to school. Two trained monitors lead organized walking caravans of up to 50 children on each route.

The key innovation of *Ciempiés* lies in its meeting point system, illustrated in Figure 1. Along each safe path, the program establishes designated meeting points where parents or caregivers can drop off and pick up children. This system reduces the coordination burden on families in several ways. Parents no longer need to accompany their children the entire way to school. Children benefit from adult supervision and peer interaction during the commute. Families gain flexibility in their daily schedules since the meeting points are strategically located to be accessible from multiple residential areas.

The program targets primary school children (ages 6-10) attending public schools. While there is an eligibility criterion requiring students to live within 700 meters of a meeting point, this criterion is not strictly enforced in practice, allowing program staff to accommodate families based on local needs and logistical considerations.

Ciempiés expanded through a staggered rollout that provides the quasi-experimental variation for this study. Starting with a pilot in 2018, and a first phase of the rollout in 2019, the program has grown steadily, reaching 53 school shifts across 9 of Bogotá’s 19 localities by 2024. Figure 6 shows the spatial distribution of the program as of 2022. The program is scheduled to expand to an additional 75 schools through 2025-2030.

The cost of the program is approximately USD 285 per child per year (at 2023 prices), which includes monitor salaries, program administration, and equipment. This cost is significantly lower than formal transportation alternatives or intensive educational interventions, making it an attractive option for resource-constrained educational systems.

2.2 Data

We combine administrative data from three institutions (two at the local and one at the national level). Our three data sources are the following:

Secretary of Mobility. From Bogotá’s Secretary of Mobility, we obtain detailed information about the [WSB](#) operation, including application and enrollment records for all participating children with their geocoded home addresses. The dataset includes daily attendance records for the walking caravans and documents which parent or caregiver dropped off and picked up each child at the meeting points.

Additionally, the Secretary of Mobility provided the universe of traffic accident reports in Bogotá from 2015-2024. These records include precise geolocation data, accident severity, victim demographics (age, gender), mode of involvement (pedestrian, passenger, driver, cyclist, motorcyclist), and timestamps. This comprehensive dataset allow us to examine effects of the program on accidents involving children of primary school age, and spillover effects to other age groups and geographic areas.

Secretary of Education. From Bogotá’s Secretary of Education, we access the SIMAT (Integrated System of Student Enrollment), an annual registry of all students enrolled in Colombian public schools from 2016-2024.⁸ This dataset includes socioeconomic information

⁸For private schools, we only have data from 2020 onwards, since before that year they were not in SIMAT.

such as ethnicity, disability status, country of origin, socioeconomic stratum⁹, and SISBEN scores. The data allows us to look at outcomes like grade progression, school switching, and dropout.

We also have access to records of alternative mobility programs and transportation subsidies from 2022 to 2024. We merge these records with SIMAT data to obtain a comprehensive picture of the transportation assistance available to households. The other mobility programs are *Al Colegio en Bici*, a program that organizes caravans of children biking to school with bicycles lent by the city government, and *Biciparceros*, which organizes groups of children (older than 14) to go to school with their own bikes, providing supervision and mechanical assistance along the way. Figure A.3 shows the shares of children in the different mobility programs (including *Ciempiés* and transportation subsidies), by deciles of distance to school.

Our learning outcomes come from the *Plataforma de Apoyo Escolar* (2015-2024), a system that the Secretary of Education provides to schools for generating school transcripts. From these transcripts we retrieve the end-of-year math and language grades, and in some cases, teacher-recorded absences and measures of parental involvement. Teachers assign grades at their discretion within each school, but they must use a national scale with four achievement categories: low, basic, medium, and high.

Finally, we also have access to novel data on school issued alerts from the *Sistema de Alertas*, which schools have to report by law to the Secretary of Education. The data covers behavioral alerts, and violence and accident reports from 2014-2024, and includes the time and description of the event, and the identity of the aggressor, disruptive kid or victim involved. We use the alerts issued for any accident that happened on the way to school to complement our analysis on the impact of *Ciempiés* on road safety outcomes.

Instituto Colombiano para la Evaluación de la Educación (Icfes). From Icfes, we obtain standardized test scores from the *Pruebas Saber* 3, 5, and 9 for 2017. This was the last year that standardized tests were applied broadly to primary school students before an interruption that lasted until 2022. After that year, only a representative sample of schools is tested.¹⁰ These data allow us to assess balance between schools that adopted *Ciempiés* at some point

⁹The socioeconomic stratum, known as *estrato* in Spanish, refers to Colombia's socioeconomic stratification system, which categorizes households based on their affluence using neighborhood and dwelling characteristics, with one corresponding to the poorest, and six to the richest households.

¹⁰The *Pruebas Saber* evaluate student competencies in math, language, natural sciences, and citizenship across 3rd, 5th, and 9th grades. The 2017 edition only administered language and math tests. Individual test scores are categorized into four performance levels: insufficient, minimum, satisfactory, and advanced. Table A.1 describes each of these categories.

between 2019 and 2024 and future adopters (never-treated group) before the program began, providing an indirect way to evaluate parallel trends.

Table 1 summarizes the characteristics of children participating in *Ciempies* from 2021-2024. Participants are typically 8 to 9 years old (corresponding to third and fourth grades). Half of them are girls. The average child in our sample comes from socioeconomic stratum 2, indicating these are vulnerable children from low-income families. The data also reveal a higher representation of Venezuelan children and victims of Colombia’s armed conflict, particularly in recent years. These groups face additional barriers to accessing education. For comparison, 7% of Bogotá’s public school students were Venezuelan, and 4% were victims of Colombia’s armed conflict in 2024, versus 13% and 8% in our sample.

Table 2 provides descriptive statistics for the walking paths from 2020-2024. The average path length is approximately 1.5-1.6 kilometers, requiring 40-42 minutes of group walking time. The program serves different school shifts, with a relatively balanced distribution between morning and afternoon shifts, plus some full-day programs.

These path characteristics highlight several important features of the program. First, the walking distances are substantial, which underscores why parents may have safety concerns and used to accompany their children to school or arrange a transportation service in the absence of the program. Second, the 40+ minute (one way) walking time represents a considerable daily time investment for families, making the supervised caravan model particularly valuable for working parents who cannot accompany children for such extended periods.

3 Empirical Strategy

To estimate the causal impact of *Ciempies* on children’s educational outcomes and road safety, we exploit the program’s staggered rollout across Bogotá’s public schools between 2019 and 2024. We implement a DiD approach using the methodology proposed by Callaway and Sant’Anna (2021), which allows us to estimate group-time Average Treatment Effect on the Treated (ATT) for cohorts of children that began participating in different years.

This method compares the evolution of outcomes for treated children in schools where *Ciempies* started operating relative to children in schools that have not yet implemented the program, but where the program expanded later or is set to expand in 2025-2030. Specifically, for each treatment cohort g and year t , we estimate:

$$ATT(g, t) = \mathbb{E}[Y_{it} - Y_{ig-1} \mid G_i = g] - \mathbb{E}[Y_{it} - Y_{ig-1} \mid G_i = g'], \text{ with } g' > t \quad (1)$$

where Y_{it} is the outcome of interest (e.g., dropout, grade repetition, switching schools or school branches, math and language grades) for student i in year t , and $G_i = g$ indicates the year cohort g in which the student started treatment. We use as controls all students in schools that have not been treated yet by time t (so cohorts $g' > t$). For the earlier cohorts, these include not-yet treated, and never-treated students, while for the last treated cohort (the one that started with *Ciempiés* in 2024), these include only never-treated students ($g = \infty$).

The main assumption we have to make for the internal validity of our estimates is the parallel trends assumption based on not-yet-treated units, which states that in the absence of treatment, average potential outcomes of the group first treated in g and the group not-yet-treated in $t > g$, would have evolved in parallel.

We compute event-study estimates by grouping treatment effects across different cohorts by the number of years that have passed since program implementation:

$$ATT_l^w = \sum_g w_g \cdot ATT(g, g + l) \quad (2)$$

where l indexes the number of years since treatment began, and w_g represents the relative size of cohort g . This aggregation enables us to examine dynamic treatment effects and assess whether the program's impact grows or fades over time.

In our main specifications, we focus on a set of educational outcomes at the individual level using SIMAT: school switching (including switching between branches or *sedes*), grade repetition, and dropout. We measure dropout as a dummy variable that takes the value of one if, by the end of school year t , the student i didn't enroll in the public school system the following year. We define grade repetition in t as being in the same grade as in $t - 1$.

To estimate the unintended effects on traffic accidents, we apply a similar DiD framework to traffic accident data but at the census tract level. We compare the number of accidents in census tracts intersecting with operating school paths to those in census tracts that intersect not-yet-treated school paths.¹¹ Additionally, we expand the not-yet-treated control

¹¹We use a catchment area of 250m around school paths to define treated versus control census tracts. Figure 7 illustrates the procedure that we follow.

group to include a never-treated control group consisting of all untreated census tracts in the *localidades* where *Ciempiés* operates.

Since our data on accidents is anonymized, we can’t establish whether the accident victim is attending any of the schools in our sample. As such, our estimates recover both the effect of the program and any spillover effects that may occur if other pedestrians or drivers change their behavior along the safe paths to school. For this reason, we complement our analysis by focusing on the school issued alerts on student road accidents, which schools need to report to the Secretary of Education. We estimate Equation 1 at the school level for all the periods before and after program implementation, and for all cohorts, and also present these results in the next section.

4 Results

Our empirical strategy provides the causal impact of the *Ciempiés* WSB program on children’s educational outcomes and road safety. Before turning to our main results, we examine whether schools that adopted *Ciempiés* at some point between 2019-2024 differed systematically from the not-yet adopters where the program will expand in the future. Table 3 reports means and standardized differences of baseline covariates measured in 2017-2018 (before any school adopted the program), including demographics, socioeconomic strata, traffic accidents on the way to school, and standardized test scores from the 2017 Saber assessments. We present both unweighted and student-weighted normalized differences.

The balance in covariates between adopters and non-adopters provides an indirect way of assessing the plausibility of the parallel trends assumption (Baker et al., 2025). If treated and control units are similar in observable variables that explain potential education outcomes and trends, this strengthens the claim that potential outcomes would have evolved in parallel in the absence of treatment. Our results in Table 3 show relatively small standardized differences between adopters and non-adopters, with most falling below 0.25 in absolute value—a commonly used threshold for meaningful imbalance (Baker et al., 2025). This evidence supports the credibility of our identification strategy and complements the graphical inspection of pre-trends that we discuss below.

4.1 School Switching

We now turn to the main results from our DiD estimation using the Callaway and Sant’Anna (2021) methodology, exploiting the staggered rollout of the program across Bogotá’s public schools between 2021 and 2024.

Figure 8 presents our dynamic estimates of the impact of *Ciempiés* on children’s educational trajectories. The corresponding estimates are reported in Table 4.

The event-study plots in Figure 8 show the dynamic *ATT* for four key outcomes: dropout, grade repetition, school switching, and switching between school branches. Each panel displays treatment effects relative to the year before program implementation, aggregated among the different cohorts, with negative time periods indicating pre-treatment periods and positive values representing post-treatment effects. Since the 2021 cohort is the only one we can observe for three years after treatment, and it is the smallest ($N = 771$), our results for $t = 3$ (and $t = 2$ for dropout) are noisy.

Panel C of Figure 8 shows that *Ciempiés* significantly reduces school switching. The effect becomes apparent in the first year after treatment, with treated students being 3.4 percentage points less likely to switch schools. Two years after treatment *Ciempiés* decreases school switching by 7.7 percentage points. Relative to the pre-treatment school switching rate of 14.4% in the control group, these represent reductions of 24% and 53%, respectively.

We find even more pronounced effects for the switching between school branches variable. Figure 8, Panel D shows that *Ciempiés* reduces branch switching by 5.4 percentage points in the first year after treatment and by 10.6 percentage points in the second year. Compared to the pre-treatment branch switching rate of 23.8% in the control group, these effects represent reductions of 23% and 45%, respectively. These large effects suggest that the program substantially increases children’s attachment to their school by reducing transportation coordination challenges, which often force families to change their children’s schools from one year to the next.

The pre-treatment coefficients in both panels show no systematic differences between treatment and control groups before program implementation, supporting the parallel trends assumption underlying our identification strategy. The growing magnitude of effects over time suggests that the benefits of the program compound as families become more accustomed to the *Ciempiés* routine and the social bonds within the walking caravans strengthen.

4.2 Grade Progression and Dropout

For grade repetition and dropout, shown in Figure 8, Panels A and B, respectively, we observe point estimates suggesting program benefits, though these effects are imprecisely estimated with large confidence intervals. Panel B shows that *Ciempiés* reduces grade repetition by 4.1 percentage points in the second year after treatment. Compared to the pre-treatment

grade repetition rate of 2.4% in the control group, this reduction is substantial, though the statistical imprecision warrants caution in interpretation.

Similarly, Panel A suggests the program reduced dropout by 5.0 percentage points two years after treatment, representing a 55% reduction from the pre-treatment dropout rate of 9.0% in the control group. This effect is also statistically imprecise, reflecting our limited power for long-term outcomes.

4.3 Results on Accidents

While the program was designed with road safety as a key component, our analysis of traffic accident data reveals more nuanced results. Figure 9 presents event-study estimates for accidents involving children 12 years old or younger in census tracts overlapping with the *Ciempiés* school paths.

Contrary to the program’s safety objectives, we find no significant impact on traffic accidents involving primary school-age children. The event-study coefficients fluctuate around zero throughout both pre- and post-treatment periods, with confidence intervals that include zero for all time periods. This null effect persists whether we use only not-yet-treated units as controls or add never-treated units to the comparison group.

Our null results on accidents do not necessarily indicate that the program lacks safety value for children. Since we can’t observe in our accident data whether the kid involved in the accident is enrolled in any of our treated schools, it is still plausible that the program may be increasing road safety for children in *Ciempiés* schools, but not for every child under 12 years old of age that walks along the treated census tracts.¹²

In an attempt to disentangle between these two, we estimate an specification at the school level, using data from school-issued accident alerts. This data contains any accident happening at the school premises or on the way to school. Figure 10 shows our event study estimates of the program’s impact on accidents occurring on the way to school. We find null effects for the first two years post-treatment, and estimates that fluctuate between 0.4 and 0.1 thereafter, though these are not statistically different from zero.

We interpret these results as indicating a lack of improvement in the safety of children’s daily commutes. This likely reflects the interaction of two opposing forces. On one hand, the program’s road safety components—including the design of safe walking routes that minimize

¹²The program may have also improved road safety for treated children, while creating other points of congestion/traffic accidents in nearby streets, offsetting the potential gains from designing safe paths to school with adult supervision.

traffic exposure and educational activities promoting traffic awareness—should reduce the risk of traffic accidents. The presence of trained monitors who can stop traffic at dangerous intersections (as illustrated in Figure A.1) should also help in this regard. On the other hand, the program simultaneously reduces adult supervision relative to the counterfactual scenario where parents individually accompany their children to school. Under the *Ciempies* model, just two trained monitors supervise groups of up to 50 children, creating a lower adult-to-child supervision ratio. This reduction in adult supervision may offset some of the program’s safety benefits. In addition, if the mechanism that drives our estimated impacts on educational outcomes is increased attendance, we would expect more children on the streets during school commuting times, increasing the probability that an accident happens.

Our results demonstrate that the *Ciempies* program successfully addresses the coordination challenges that prevent many families from maintaining consistent school attendance patterns. The program’s most significant impacts are evident in measures of school retention and attachment; students are substantially less likely to switch schools or school branches. These effects are important, as dissatisfaction with public schools has increased post-pandemic, with many parents switching to substitutes of public schools (e.g., Francis and Goodman, 2025, in the U.S.). Our estimated effects grow stronger over time, suggesting that the benefits compound as families integrate the program into their routines.

While we do not find evidence of improved road safety for children, the program’s primary value lies in its ability to reduce the logistical burden on families while maintaining adequate supervision of children’s school commutes. The effects on educational continuity, achieved at a cost of just USD 285 per child per year, position *Ciempies* as a highly cost-effective intervention for improving school attendance in urban settings.

These findings have important implications for education policy in similar contexts. The results suggest that relatively simple, community-based approaches to addressing transportation barriers can generate meaningful improvements in educational outcomes without requiring large-scale infrastructure investments or intensive individual interventions. The success of *Ciempies* points to the potential value of addressing the practical, day-to-day challenges that families face in accessing education, rather than focusing exclusively on supply-side improvements or demand-side incentives.

5 Conclusion

This paper shows that relatively simple, community-based interventions can meaningfully improve educational outcomes by addressing the practical coordination challenges that fam-

ilies face when accessing education. The *Ciempiés* program in Bogotá successfully reduced school and branch switching by up to 7.7 and 10.6 percentage points, respectively, after two years in the program. These effects represent 53% and 45% reductions from pre-treatment rates in control schools, indicating that the program significantly increases children’s attachment to their schools and reduces the disruptions that often force families to seek alternative schooling options.

While we find suggestive evidence of reductions in grade repetition and dropout, these effects are imprecisely estimated, likely reflecting our limited sample size for long-term impacts. Importantly, we do not find evidence that the program improved road safety for children, suggesting that the primary value of *Ciempiés* lies in addressing coordination and logistical barriers rather than direct safety improvements. At a cost of just USD 285 per child per year, the program is a highly cost-effective way to improve school retention compared to intensive, individualized interventions or formal transportation alternatives. These findings highlight the potential of place-based policies that improve educational opportunities within existing communities without requiring families to relocate or children to change schools, offering a promising model for addressing educational inequality in other urban contexts across LMICs.

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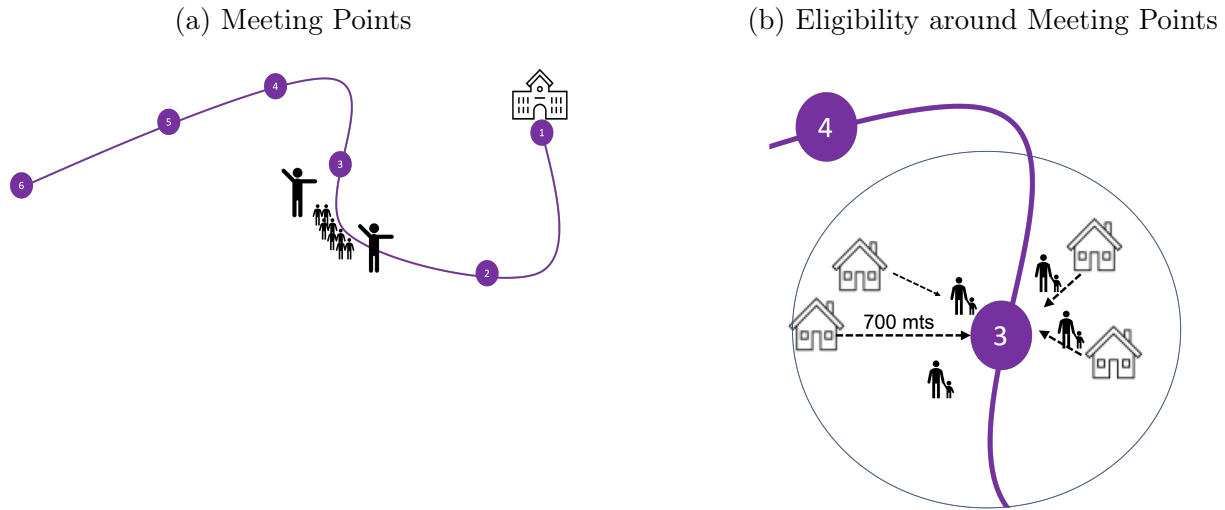
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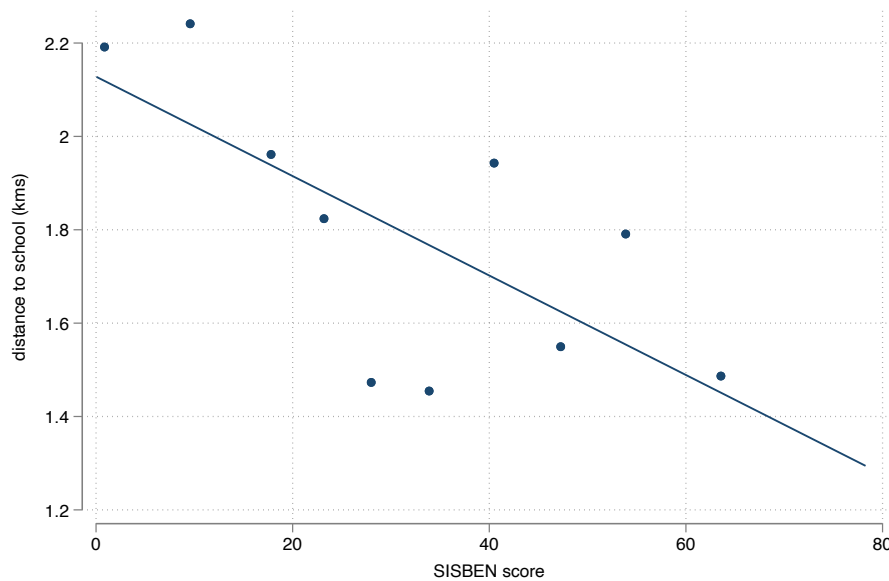
Figures and Tables

Figure 1: WSC Operation



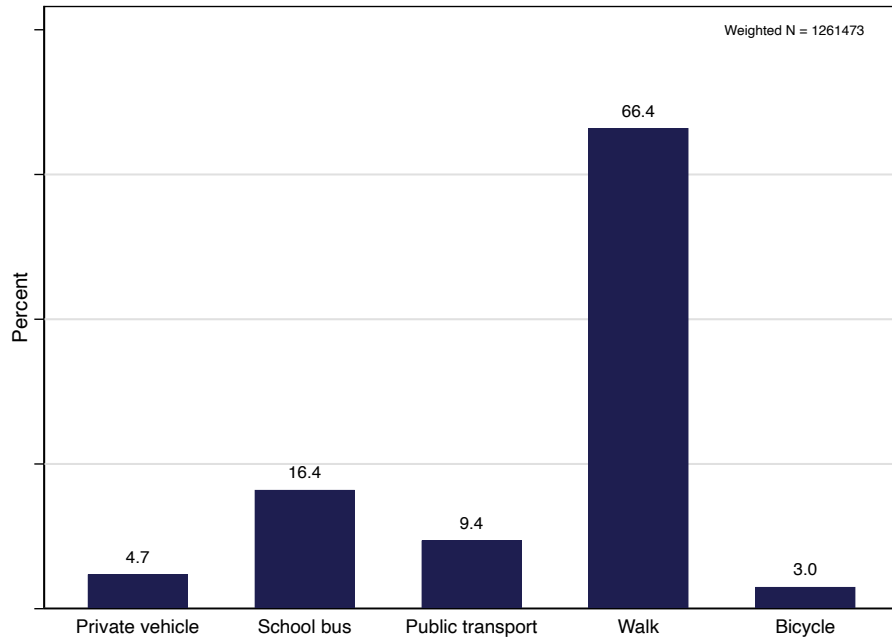
Notes: This figure shows a depiction of the WSC operation. Panel A shows the meeting points along the safe path to school. Panel B shows how a meeting point is used by the households living close by to drop and pick up children. Although there is an eligibility criterion on paper that says that students need to live within a 700m radius of any meeting point, this criterion is not enforced in practice.

Figure 2: Binscatter Regression of Distance to School on SISBEN Score



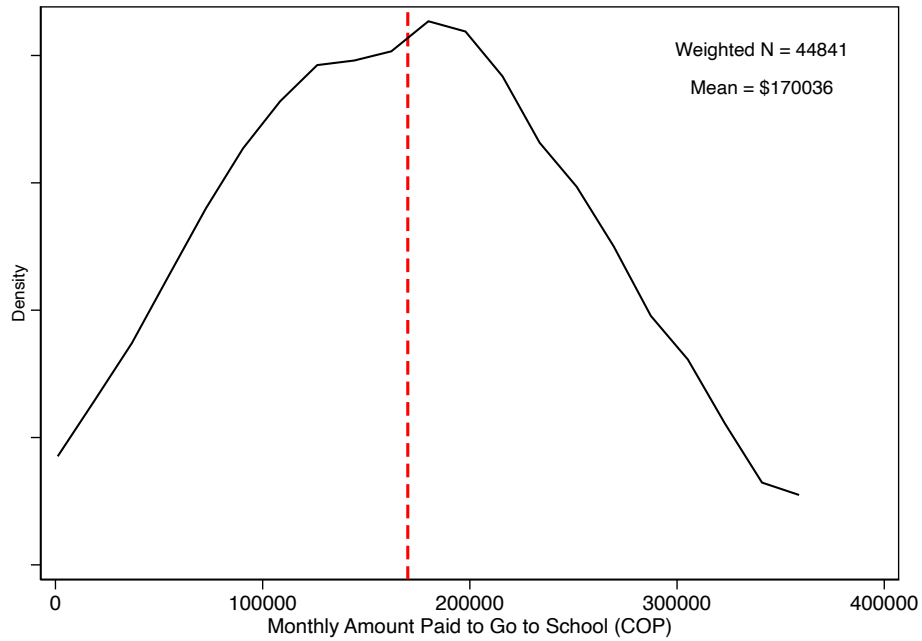
Notes: This figure shows the relationship between distance to school and SISBEN score, a household vulnerability index used to target social programs in Colombia. The sample corresponds to students in *Ciempies* schools in 2021. The SISBEN score ranges from 0 to 100, with lower scores indicating higher vulnerability, and a score of 0 indicating extreme poverty.

Figure 3: School Transportation Modes in Urban Bogotá



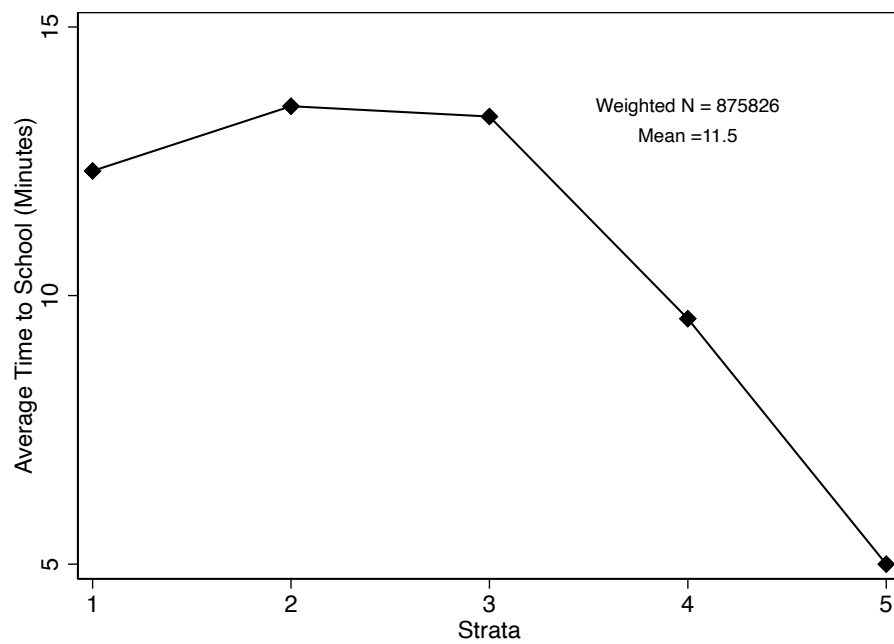
Notes: This figure shows the different transportation modes for children in public and private schools in Bogotá. The data corresponds to Colombia's 2024 [QLS](#).

Figure 4: Commuting Costs to Schools



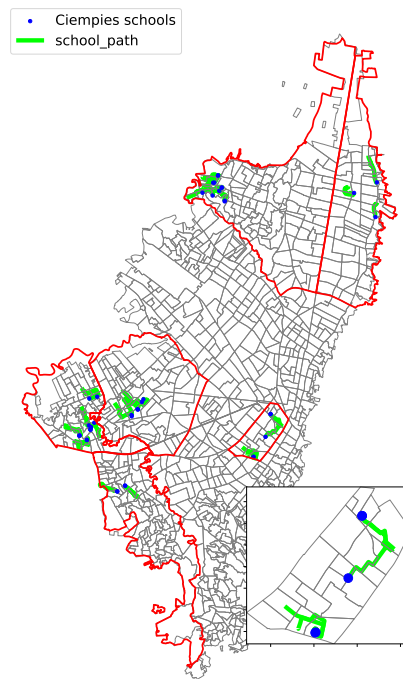
Notes: This figure shows the distribution of monthly commuting costs to school for the households that report paying for a transportation service. The dashed red line corresponds to the mean. The data corresponds to Colombia's 2024 [QLS](#). All values are in COP.

Figure 5: Commuting Time to School by Socioeconomic Stratum



Notes: This figure shows the relationship between commuting time to school and socioeconomic stratum for children that commute to school by bike or on foot. The data corresponds to Colombia's 2024 [QLS](#).

Figure 6: Ciempiés Schools and Paths in 2022



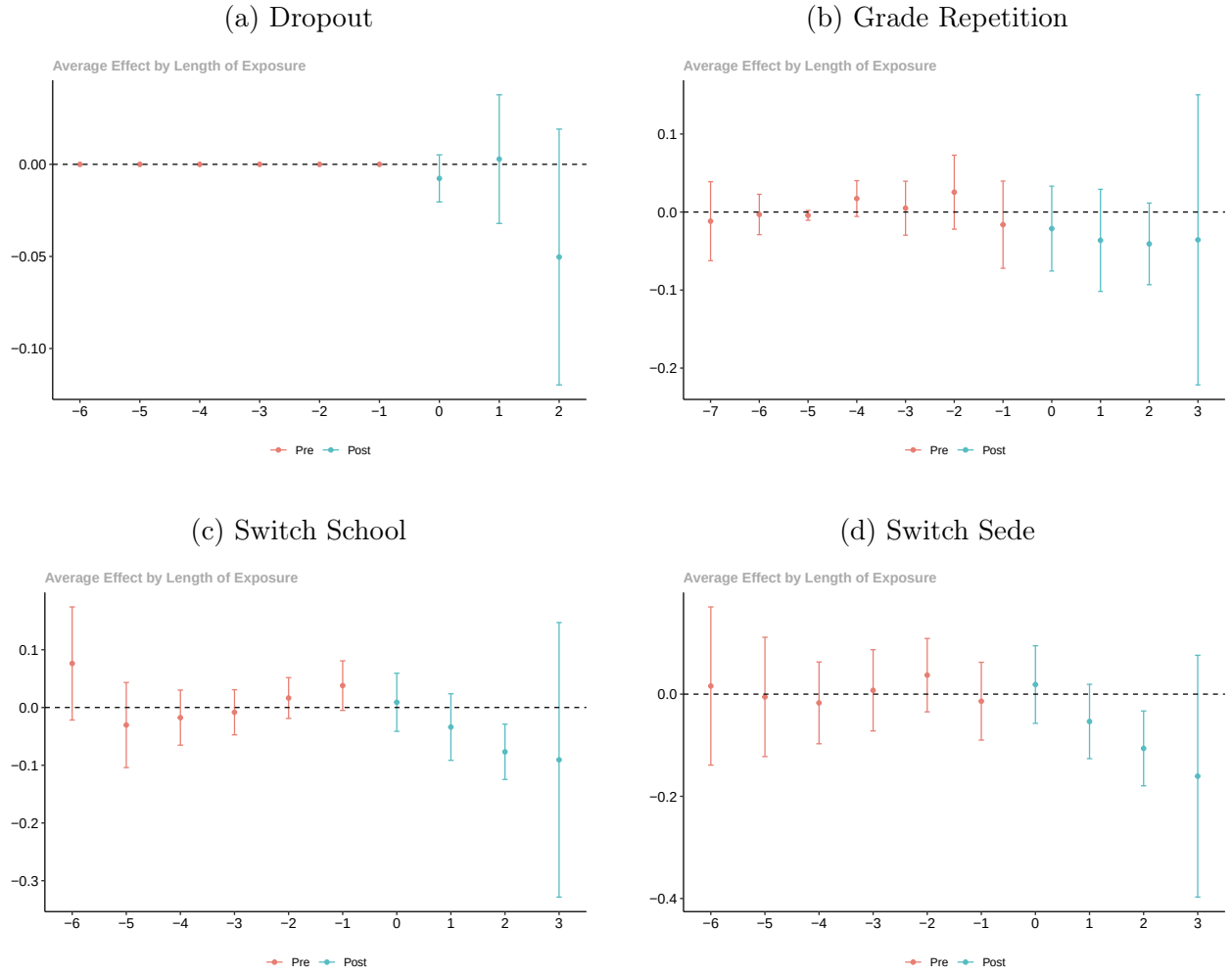
Notes: This figure shows the spatial distribution of *Ciempies* schools and paths as 2022. As of that year, the program was only operating in 6 of the 19 *localidades* in Bogotá (circled) in red, but has expanded its operations to 9 as of 2024.

Figure 7: Defining Treatment and Control Census Tracts for Traffic Accident Analysis



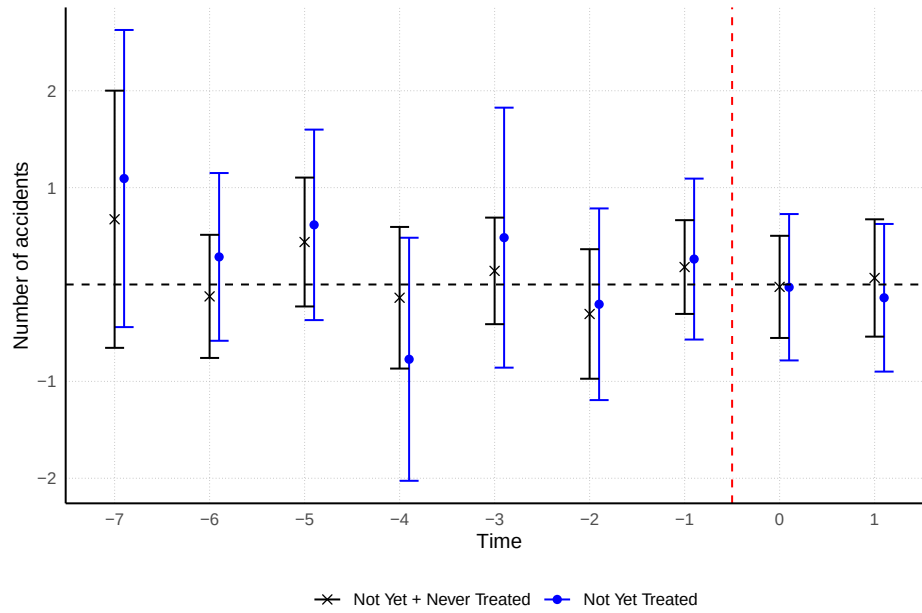
Notes: This figure shows the procedure to define treated versus control census tracts using catchment areas around school paths. We define a treated census tract if its area intersects the catchment area of a school path in at least 50%.

Figure 8: Impact of Ciempiés on School Trajectories of Children



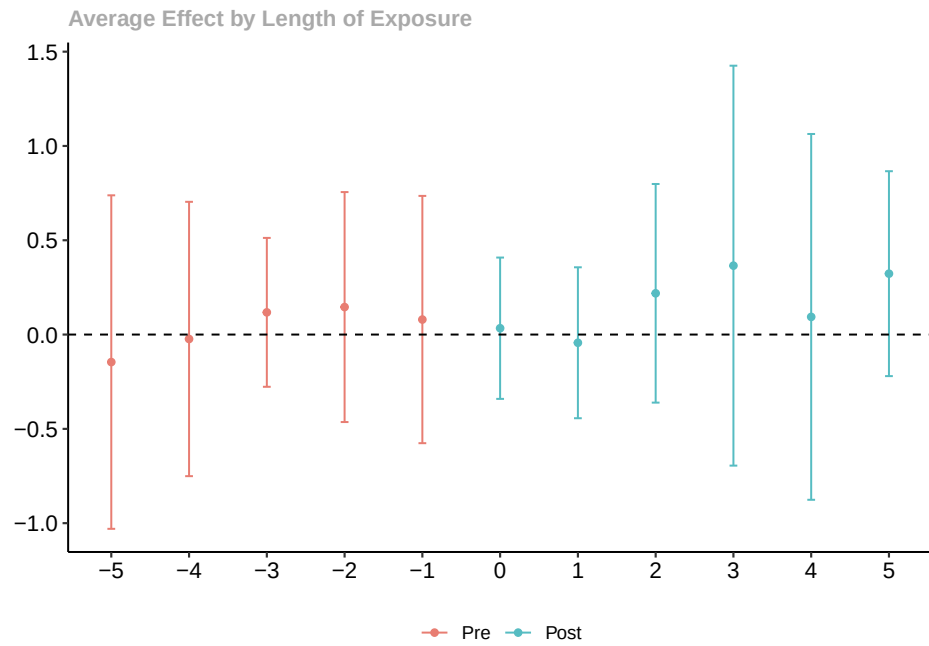
Notes: This figure shows the event-study aggregation of the estimated ATT for dropout, grade repetition, school switching and sede switching. The control group are the not yet treated students and the never treated students in school where the program is set to expand. 95% confidence intervals reported.

Figure 9: Results for Accidents Involving Children ≤ 12 Years Old



Notes: This figure shows the event-study aggregation of the estimated ATTs on census tract accidents involving children less than 12 years old. The estimates in blue only use not yet treated units in the control group, while the estimates in black use both not yet treated and never treated units. 95% confidence intervals reported.

Figure 10: Results for Road Accidents at the School Level



Notes: This figure shows the event-study aggregation of the estimated impacts of the program on school level accidents, relative to the year before the program. The data comes from the *Sistema de Alertas*. We compare earlier cohorts of entrants with not-yet-treated and never-treated schools. 95% confidence intervals reported.

Table 1: Descriptive Statistics for Ciempiés Children, 2021–2024

	2021	2022	2023	2024
Age	8.741 (1.996)	9.324 (2.421)	9.359 (2.505)	9.395 (2.273)
Grade	3.276 (1.839)	3.778 (2.272)	3.775 (2.336)	3.785 (2.089)
Share female	0.529 (0.499)	0.529 (0.499)	0.513 (0.500)	0.505 (0.500)
Socioeconomic stratum	2.069 (1.076)	1.982 (1.311)	2.051 (1.446)	2.113 (1.503)
Share victim conflict	0.035 (0.184)	0.042 (0.200)	0.090 (0.286)	0.077 (0.267)
Share Venezuelan	0.095 (0.293)	0.122 (0.327)	0.150 (0.358)	0.133 (0.339)
N	771	2139	2193	2820

Notes: This table shows means and standard deviations of student characteristics in our sample of Ciempies children, for the years 2021–2024.

Table 2: Descriptive Statistics for School Paths, 2020–2024

	2020	2021	2022	2023	2024
Length (km)	1.486 (0.392)	1.643 (0.481)	1.648 (0.354)	1.570 (0.361)	1.513 (0.305)
Walking time (min)	38.286 (9.736)	41.389 (8.558)	41.500 (8.767)	41.890 (7.176)	40.915 (6.452)
Share morning shift	0.571 (0.514)	0.481 (0.509)	0.475 (0.506)	0.439 (0.502)	0.434 (0.500)
Share afternoon shift	0.214 (0.426)	0.407 (0.501)	0.425 (0.501)	0.439 (0.502)	0.434 (0.500)
Share full shift	0.214 (0.426)	0.111 (0.320)	0.100 (0.304)	0.122 (0.331)	0.132 (0.342)
N	14	27	40	41	53

Notes: This table shows means and standard deviations of school path characteristics, for the years 2020–2024. Length is measured in kilometers and walking time in minutes. Shift shares represent the proportion of paths in each shift (morning, afternoon, or full-day).

Table 3: Covariate Balance Statistics

variable	Unweighted			Weighted		
	Non-Adopt	Adopt	Norm. Diff.	Non-Adopt	Adopt	Norm. Diff.
Gender	0.482	0.490	0.103	0.503	0.495	-0.080
Stratum	1.933	2.007	0.270	1.907	1.984	0.285
Number of accidents	0.313	0.158	-0.259	0.575	0.202	-0.511
Insufficient (math)	0.453	0.394	-0.516	0.443	0.392	-0.488
Minimum (math)	0.312	0.344	0.483	0.314	0.337	0.405
Satisfactory (math)	0.151	0.172	0.453	0.154	0.176	0.494
Advanced (math)	0.084	0.090	0.120	0.089	0.095	0.125
Insufficient (language)	0.119	0.108	-0.189	0.107	0.101	-0.120
Minimum (language)	0.441	0.421	-0.288	0.431	0.420	-0.165
Satisfactory (language)	0.321	0.339	0.256	0.331	0.337	0.107
Advanced (language)	0.118	0.131	0.235	0.132	0.142	0.186

Notes: This table shows how the 43 school branches that adopted Ciempiés between 2019 and 2024 compare to the 75 branches that did not adopt it by 2024. We present the means and standardized differences for each variable by adoption status, both unweighted and weighted by the number of primary school students in each branch. All variables are measured as shares, except for stratum, which is measured on a scale of 1–6, and number of accidents. All variables are measured in 2018 (the year before the first cohort adopted Ciempiés), with the exception of Saber variables, which are measured in 2017. This was the last time that standardized test scores were applied to 5th grade children in Colombia.

Table 4: Dynamic ATTs by Outcome

	Repetition	Dropout	Switch School	Switch Sede
ATT(-7)	-0.012 (0.018)			
ATT(-6)	-0.003 (0.009)	0.000	0.076 (0.037)	0.016 (0.054)
ATT(-5)	-0.004 (0.004)	0.000	-0.030 (0.029)	-0.005 (0.044)
ATT(-4)	0.017 (0.008)	0.000	-0.018 (0.018)	-0.017 (0.028)
ATT(-3)	0.005 (0.012)	0.000	-0.008 (0.015)	0.007 (0.029)
ATT(-2)	0.025 (0.017)	0.000	0.016 (0.014)	0.037 (0.027)
ATT(-1)	-0.016 (0.020)	0.000	0.038 (0.018)	-0.014 (0.027)
ATT(0)	-0.021 (0.019)	-0.008 (0.006)	0.009 (0.019)	0.019 (0.026)
ATT(1)	-0.036 (0.022)	0.003 (0.017)	-0.034 (0.020)	-0.054 (0.026)
ATT(2)	-0.041 (0.019)	-0.050 (0.046)	-0.077 (0.018)	-0.106 (0.027)
ATT(3)	-0.036 (0.068)		-0.091 (0.090)	-0.161 (0.085)
N	42 121	53 888	42 121	42 121
Time Periods	9	8	8	8
Cohorts	4	3	4	4

Notes: Standard errors in parentheses. ATT estimates from dynamic difference-in-differences model using the Callaway and Sant’Anna (2021) method. Negative (positive) numbers indicate periods before (after) treatment. Control group consists of never-treated + not-yet-treated units.

“Safe Paths to School: Evidence from Bogotá’s Walking to School Caravans”

Online Appendix

Rafael Hernández-Pachón
Isabela Munevar

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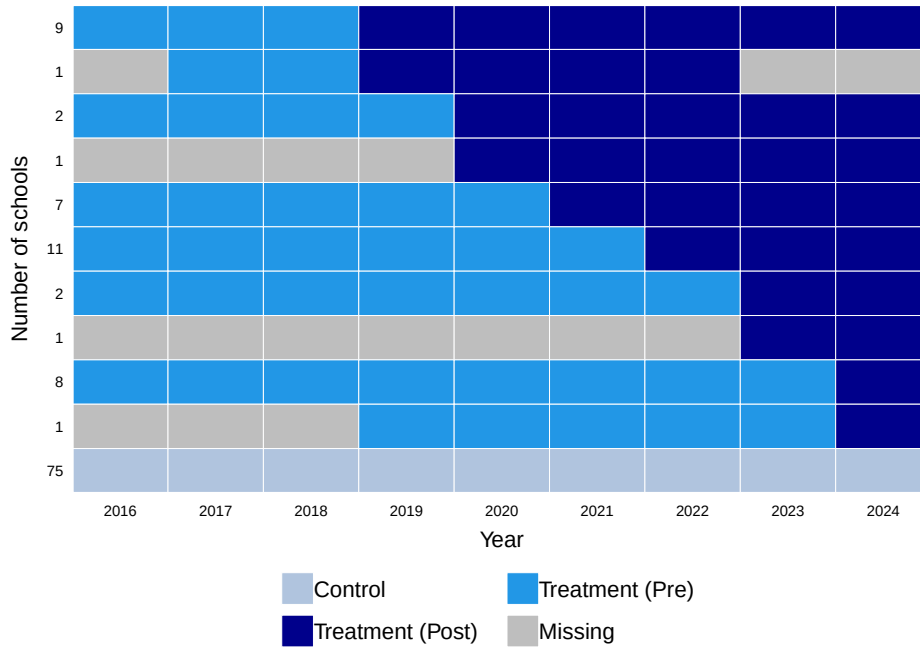
A Figures and Tables

Figure A.1: Example of Walking to School Caravan



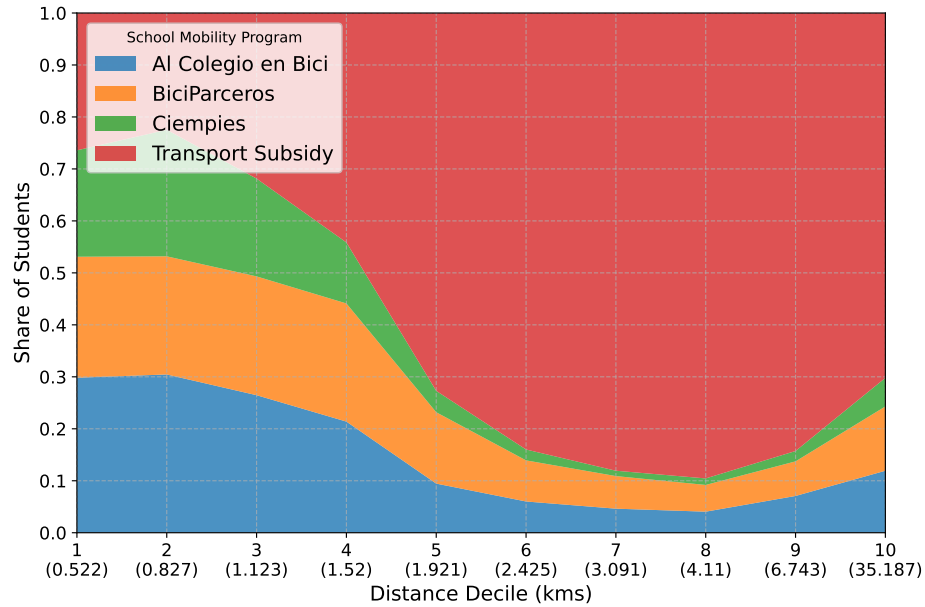
Notes: This figure shows photos from two Walking to School Caravans. The caravans are supervised by two adults, who ensure a safe commute to school. In the left panel we can see that these monitors often stop traffic for children to be able to cross the road.

Figure A.2: Treatment Rollout



Notes: This figure shows the treatment rollout from 2016 to 2024, marking the pre and post treatment periods for each cohort. The missing periods correspond to schools that opened or closed at some point during the sample period, and thus are not observed throughout the 9 years.

Figure A.3: Shares of Mobility Programs by Distance to School in 2024



Notes: This figure shows the shares of students in the different mobility programs by decile of distance to school. The max value of the distance to school in each decile is reported in parenthesis.

Table A.1: Description of Saber 5 Performance Levels

Level	Description
Advanced	Shows outstanding performance in the expected competencies for the evaluated subject area and grade.
Satisfactory	Demonstrates adequate performance in the required competencies for the subject area and grade. This is the level of performance that all or most students should reach.
Minimum	Shows minimum performance in the required competencies for the subject area and grade.
Insufficient	Does not surpass the most basic questions in the test.

Notes: This table shows the performance levels corresponding to the Saber 5 assessment scale. Saber 5 is a standardized test administered to fifth-grade students in Colombia to evaluate competencies in math and language.

B List of Acronyms

LMICs Low- and Middle-Income Countries	2
WSB Walking to School Buses	4
WSC Walking to School Caravans	3
DiD Difference-in-Differences	3
CCT Conditional Cash Transfers	2
ATT Average Treatment Effect on the Treated	10
QLS Quality of Life Survey	4